MATHEMATICAL OPTIMIZATION: LAB

Abstract. Tasks for the LAB

1. LAB 1

1. Install Julia + Jupyter Lab + JuMP solver framework + SCS solver and work through the linear programming Jump tutorial https://jump.dev/JuMP.jl/stable/tutorials/getting_started/getting_started_with_JuMP/

   Apart from Julia itself (and perhaps jupyter lab) you should be able to install most of this through the Julia package manager.

2. LAB 2

2. Check *strong duality* and *complementary slackness* for numerical solutions.
3. Dualize an infeasible LP and obtain infeasibility certificate

3. LAB 3

4. Impement LP relaxations for maximum perfect matching and vertex cover.
5. Implement the relaxations FSTAB, QSTAB for the stable set polytope.

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